ZS-A20

The Atomic Interface: A Linear-Cosmological Continuum Closure Theorem for the Z-Spin Boundary Dust — Parent-Action Equivalence to Brown Dust, Single-Trace Elimination of μ-Universality, Single-Clock Adiabaticity, and Executable Programs for a Custom-CLASS Species and the Planck/ACT/DESI Likelihood

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Verification: 76/76 PASS (92 sub-checks in released script) | C1 continuum (NEW v2.0): a first-order parent action reduces to the A19 antisymmetric-current form and yields Brown dust ($T^{\mu\nu} = \rho\, u^\mu u^\nu$, $p = 0$, irrotational, $c_s^2=0$); a single-trace matrix current makes $\rho_c/\rho_b = \mathrm{Tr}P_c/\mathrm{Tr}P_b = 16/3$ STRUCTURAL (μ-universality eliminated); a shared clock gives one physical scalar and $S_{cb} = 0$. ⇒ linear-cosmological continuum DERIVED, conditional only on C1-ID-global (charge quantization $N_i = 32, 6$ from H^3) | C2 ZHCS species VERIFIED-IN-EINSTEIN-FLUID-SOLVER; custom-CLASS species (§8.5 spec, gates B1–B6) = TARGET | C3 AT-6a PASS (parameter-surrogate); AT-6b (clik; §11.2 program ZS-C, gates L0–L10) = TARGET | m_e branch per ZS-M20 ($\sigma_1/\sigma_3 = 3477$ EXACT) | Zero new fitted Z-Spin shape parameters

§0. Abstract

Z-Spin (ZS-A20) treats recombination with a conserved boundary-tension dust (the ZHCS construction of ZS-A19) and tests a *geometry-restricted* cosmology in which the dark-matter and baryon abundances are fixed dimensionless polyhedral fractions $\Omega_{cdm} = 32/121$ and $\Omega_b = 6/121$ ($\Omega_c/\Omega_b = 16/3$), the tilt is $n_s = 0.9674$, and the electron mass is time-invariant. No CMB shape parameter is fitted: h is fixed by the acoustic scale θ_s and the standard amplitude pair $\{\tau, A_s\}$ is marginalized, so the model carries zero new fitted shape parameters. Run through CLASS+HyRec, the zero-parameter spectrum matches a Planck-2018 reference at Planck noise ($\Delta\chi^2 \approx 0$; cosmic-variance-limited residual $\Delta\chi^2 \approx 204$), with $z_{rec} = 1088.8$.

This version (v2.0) closes the continuum chain at the linear-cosmological level and lays out executable programs for the two remaining computational targets. (1) Continuum (NEW): a first-order parent action $S = -\mu_Z\int\sqrt{(-g_{\mu\nu}J^\mu J^\nu)} + \int J^\mu\partial_\mu T$, with the conserved current written as $J^\mu = \partial_\nu\Sigma_Z^{\mu\nu}$, reduces to the A19 antisymmetric-current form and, on variation, gives Brown dust ($T^{\mu\nu} = \rho\, u^\mu u^\nu$, $p = 0$, irrotational $\nabla_{[\mu}u_{\nu]} = 0$, $c_s^2 = 0$). A **single-trace matrix current** with orthogonal projectors (rank 32, 6) makes $\rho_c/\rho_b = \mathrm{Tr}P_c/\mathrm{Tr}P_b = 16/3$ a structural identity — eliminating μ-universality as an assumption — and a shared clock yields exactly one physical scalar with vanishing relative entropy $S_{cb} = 0$. The A19-action→linear dust link is thereby promoted from DERIVED-CONDITIONAL on three conditions to **DERIVED, conditional only on the global charge quantization** (C1-ID-global: $N_i = 32, 6$ as integer fluxes of $H^3(M_Z, \mathbb{Z})$), the sole remaining open step. **(2) Boltzmann:** §8.5 gives the exact custom-CLASS species patch (background/perturbation/IC modules, both gauges) and the equivalence gates B1–B6; the in-CLASS run that would raise VERIFIED-IN-EINSTEIN-FLUID-SOLVER to VERIFIED-IN-FULL-BOLTZMANN remains TARGET. **(3) Likelihood:** §11.2 pre-registers the geometry-restricted model ZS-C (fix $\Omega_b = 6/121$, $\Omega_{zhcs} = 32/121$, $n_s = 0.9674$, $m_e(z)$=const; sample h, τ, A_s

+ nuisance) and the run plan P0–P2 / A1–A3 / D1 with gates L0–L10 and an empirical-p / ΔAIC decision rule for AT-6b; the parameter-surrogate AT-6a still PASSES ($\Delta\chi^2 = 1.65$). ZS-M20 keeps $\sigma_1/\sigma_3 = 3477$ EXACT (within-branch σ_{th} NNLO-limited; 6.3σ/0.34σ retracted; lab favors H2 by magnitude). Zero new fitted Z-Spin shape parameters.

§0.1 Epistemic Status Legend

Table 0. Status tags used in this paper (uppercase, bold).

STATUS	DEFINITION
LOCKED	Core constant fixed upstream (A=35/437, Q=11, (Z,X,Y)=(2,3,6)); no downstream paper may modify.
PROVEN	Theorem from standard mathematics or corpus definitions; machine-verifiable.
IMPORTED-PROVEN	Theorem proven in the external literature; its Z-Spin use is a separate identification.
DERIVED	Follows from the Z-Spin action plus PROVEN inputs; zero free parameters beyond A.
DERIVED-CONDITIONAL	Derived, conditional on one or more named identifications not yet established.
COMPUTED	Numerical result produced and reproducible here (released script), not the full external pipeline.
COMPUTED code-consistency	A solver/implementation check under stated antecedents; not a physics validation.
VERIFIED-IN-EINSTEIN-FLUID-SOLVER	A species/equation set evolved inside a self-consistent Einstein-fluid (linear-scalar) integration — radiation + dust + one metric potential, not a full photon/neutrino Boltzmann hierarchy — reproducing the reference; because equal equations and ICs give equality by construction, the test shows solver sensitivity (controls diverge), not a new observational prediction.
VERIFIED-IN-FULL-BOLTZMANN	A custom species evolved inside the full CLASS Einstein–Boltzmann code (photon/neutrino multipoles, recombination source, lensing), passing the B1–B6 equivalence gates. Not yet achieved (TARGET): the compiled in-CLASS run of §8.5.
ILLUSTRATIVE-FORECAST	A what-if number under an assumed sensitivity; not a dedicated Fisher/MCMC forecast.
COUNTERFACTUAL-SENSITIVITY	A response template for a parameter shift the model does NOT predict (here, varying-m_e).
TARGET-COMPUTATION	A well-posed computation not yet executed (here: the full clik MCMC).
OPEN	A sharply stated, falsifiable question current corpus tools cannot settle.
RETRACTED	A prior claim withdrawn with documented reason (here: v1.2 §9 branch-by-cosmology).

§1. Introduction

ZS-A19 replaced the recombination dark-matter carrier with a conserved Brown–Kuchař boundary-tension dust (the ZHCS construction: Z-Spin Holographic Cold-Sector dust), fixing $\Omega_{cdm} = 32/121$ and reproducing the third acoustic-peak height in a control run assuming standard CDM transfer; the full transfer was OPEN (gate BRL-5). v1.0–v1.2 reframed the recombination floor as the QED quantity $B_H(0) = \frac{1}{2}\mu_{ep}c^2\alpha_{EM}(0)^2$, explained the matching lepton branch as a spectral fact (§3), and gave Saha-level branch differentials. This version executes the computable links with CLASS+HyRec, derives the dust stress tensor from the A19 action, and — most importantly — corrects which measurement selects the H1/H2 branch.

§2. Locked Inputs

Table 1. Inherited inputs. Ω are dimensionless fractions; ω = Ω h^2. Zero new constants introduced.

Quantity	Value	Source	Status
A; Q; (Z,X,Y)	35/437; 11; (2,3,6)	ZS-F2, ZS-F5	LOCKED / PROVEN
√(Y/X) ; R_{spec}	√2=1.41421 ; (5−φ)/(4−φ)=1.41982	ZS-F5 ; ZS-M11 §9.5.7	PROVEN
m_e (H1 / H2)	0.509089 / 0.511102 MeV	ZS-S9 Cor.I (m_τ/3477)	DERIVED-CONDITIONAL
lab m_e	0.51099895 MeV (ppb)	CODATA / Penning trap	IMPORTED-PROVEN
Ω_{cdm} ; Ω_b	32/121 ; 6/121 (Ω_c/Ω_b=16/3)	ZS-A19 ; ZS-A4/A6	DERIVED-CONDITIONAL / DERIVED
n_s	0.9674	ZS-U1 §4.2	DERIVED
Brown–Kuchař dust EOS	w=0, c□²=0, $\rho \propto a^{-3}$	imported	IMPORTED-PROVEN

§3. The Spectral Electron-Mass Branch (retained, condensed)

ZS-S8 gives two lepton-mass routes differing only in one O(1) factor: H1 uses the register ratio √(Y/X) = √2, H2 the spectral eigenvalue ratio R_{spec} = (5−φ)/(4−φ) of $L_Y|_{\rho_2}$ (ZS-M11 §9.5.7, PROVEN; (5−φ)(4+φ)=19=denom δ_X). Their ratio $R_{spec}/\sqrt{2} - 1$ = +0.397% is the H1→H2 shift in m_τ, hence in $m_e = m_\tau/3477$ (0.509089 → 0.511102 MeV) and in $B_H(0)$ (13.548 → 13.601 eV). The corpus 500k anti-numerology Monte Carlo gives H1 p = 0.78% and H2 p = 0.025% (H2 ≈30× rarer), independent of any hydrogen or CMB datum. The branch-forcing theorem (gate F-mTau.6) is OPEN.

§4. The Atomic Interface Proposition (condensed)

Given $\alpha_Z(0)$, the pole masses, and nuclear inputs, the standard NRQED/QED rates and the HyRec effective-multilevel-atom reduction determine $x_e(z)$, g(z), and the damping scale with no free parameter. Per-component status: $\alpha_Z(0)$ DERIVED-CONDITIONAL on ZS-M8 c_4; m_e^{poee} the bottleneck; $m_p = 2\pi\Lambda_{QCD}$ DERIVED-CONDITIONAL; Y_p DERIVED-CONDITIONAL from η_B + nuclear rates + τ_n; nuclear radii external measured; T_{CMB} a retained external anchor. The QED dependence on constants is IMPORTED-PROVEN; the executed CLASS+HyRec run (§8) now provides the actual $x_e(z)$ and spectra at the Z-Spin electron masses.

§5. The Transfer Engine: Action-Level Origin and the Isomorphism

The Brown–Kuchař reference fluid $S = -½\int\sqrt{(-g)}\, M[g^{\mu\nu}\partial_\mu T\partial_\nu T+1]$ varies to a pressureless perfect fluid (p=0, c_s^2=0, σ=0, $\rho \propto a^{-3}$). Its linear scalar perturbations δ' = −θ+3Φ', θ' = −(a'/a)θ+k^2Ψ are identical to standard CDM (Ma–Bertschinger conformal-Newtonian convention; the +3Φ' sign corrects a v1.3 typo, with no numerical impact since the §7 toy uses Φ'=0). Weinberg 2003 supplies the *existence* of the adiabatic mode (IMPORTED-PROVEN); *occupation* (passivity, adiabaticity) is DERIVED-CONDITIONAL on condition C3 (single-clock occupation ∧ no primordial isocurvature ∧ no continuously-active source). The A18 smeared hump **is consistent with and helps diagnose** anisotropic stress, not "precisely caused" by it. Status: Boltzmann Isomorphism

PROVEN-CONDITIONAL under perfect-dust assumptions; the A19-action→those-assumptions link is addressed in §9.

§6. Companion Scripts and Reproducibility (item 1)

All numbers in §7–§8 are produced by released Python scripts, re-run for this version. **Environment:** Python 3.12, numpy, scipy, matplotlib, and CLASS v3 via the classy wrapper (bundling HyRec). **Files:** compute_a20.py (perturbation ODE isomorphism + control; Saha cross-check; analytic error budget), class_a20.py (CLASS+HyRec runs for fiducial/H1/H2; z_{rec}, r_s, $100\theta_s$; H_0-at-fixed-θ refit; branch χ^2 with cosmic-variance covariance), and fig_a20_v13.py (Figure 1). **Added in v1.4–v1.6:** class_a20_pathA.py (Path-A absolute-Ω lock: Ω_c=32/121, Ω_b=6/121 held, h fit to θ_s, ω_i=Ω_i h², z_rec, r_s, Figure 2a inputs and the $\Delta\chi^2$ values), fig_a20_v14.py (Figure 2), minicmb_a20.py (the §8.4 self-consistent Newtonian-gauge Einstein-fluid integrator: radiation+dust+Φ, ZHCS vs CDM vs imperfect/wrong-sign controls), fisher_a20.py (the §11.1 CLASS-derivative Fisher matrix over {ω_b,ω_c,h,n_s,τ,ln10^{10}A_s}, saving fisher_F.npy), and fig_a20_v15.py (Figure 3). Each script is deterministic; CLASS settings are output='tCl,pCl,lCl', lensing='yes', l_max=2500, and the varying_fundamental_constants='instantaneous' module with varying_me set to the branch ratio. Added in v2.0: continuum_a20.py (the §9.1–§9.5 parent-action closure: timelike normalization, clock-induced irrotationality, antisymmetric-potential conservation, single-trace projector 32:6, shared-clock S_cb=0 and reduced kinetic rank 1). The consolidated suite zs_a20_verify_v2_0.py runs 92 sub-checks (categories A–N; K/L/N cover C1/C2 without CLASS, M loads fisher_F.npy). The released bundle lets an external reader reproduce every COMPUTED value below.

§7. Dust≡CDM Perturbation Equality — Code-Consistency Check

Integrating the §5 equations for a CDM species, a perfect dust ($w=c_s^2=\sigma=0$), and an imperfect dust ($c_s^2=0.01$) under identical metric source and initial data (conformal-Newtonian gauge, mode k = 0.2 in units where the conformal Hubble rate is $a'/a = 2/\eta$ on a matter-dominated toy background $a \propto \eta^2$, constant potentials $\Psi = \Phi = 10^{-4}$, initial $\delta_0 = 10^{-5}$, $\theta_0 = 0$, integrated $\eta \in [0.1, 50]$, scipy LSODA rtol = 10^{-11}, atol = 10^{-14}): $\max|\delta_{dust}-\delta_{CDM}| = 0$ to machine precision, while the imperfect dust deviates by 7.4% in relative δ at η = 50. **Honest status (per review):** because §5 already shows the two species share the same equations, this is a **COMPUTED code-consistency verification under the perfect-dust antecedents** — it confirms the solver and distinguishes the $c_s^2\neq0$ control, but it is not by itself a validation of the A19 boundary action. The arbitrary $c_s^2=0.01$ control is illustrative only.

§8. Computed Recombination and CMB Spectra with CLASS+HyRec (items 2, 3)

Replacing the v1.2 Saha estimates, we run CLASS+HyRec for three electron masses — fiducial ($m_e/m_{e,0}=1$), H1 (0.996262), H2 (1.000202) — at otherwise fixed Planck-2018 parameters, producing lensed TT/TE/EE/lensing spectra and derived recombination quantities.

Table 2. CLASS+HyRec recombination outputs (real; replaces the v1.2 Saha z_* = 1378). Absolute z_rec is now correct (≈1089); the branch differentials confirm the v1.2 Saha signs and magnitudes.

Quantity	fiducial	H1 (√2)	H2 (R_spec)
z_{rec}	1088.78	1084.65	1089.00
r_s [Mpc]	144.55	144.91	144.53
100 θ_s	1.04037	1.04303	1.04023

H1→H2 differentials (CLASS+HyRec): Δz_{rec} = +0.402%, Δr_s = −0.263% — the same sign and magnitude as the v1.2 Saha result (+0.40%, −0.27%), now confirmed with the full recombination history. Holding $100\theta_s$ fixed by refitting h gives h = 0.683, i.e. ΔH_0 = +0.94 km/s/Mpc (H2 higher), consistent in sign with the m_e–H_0 degeneracy $\Delta m_e/m_e \approx \frac{1}{2}\Delta H_0/H_0$ (Toda–Seto 2025). The larger value here (1.4% vs the ≈0.8% of the multi-parameter relation) reflects that only h was refit; the realistic shift is bracketed by the two.

§8.1 Branch χ² and the degeneracy

From the actual spectra, the cosmic-variance-limited TT/TE/EE χ² between H1 and H2 (to ℓ = 2500, f_{sky}=1) is 3145 raw — but this is almost entirely a coherent acoustic-scale shift: refitting only H_0 reduces it to 37, and the full marginalization over (h, ω_b, ω_c, n_s, A_s, τ) plus BAO that the published analyses perform collapses it to the sub-σ level. Using the fully-marginalized real-data sensitivities, the 0.394% branch separation is 0.86σ (Δχ² = 0.73) against ACT DR6 + DESI DR2 (σ_{me} = 0.46%), 0.70σ against P-ACT-LBS (0.56%), and 0.59σ against Planck 2018 + BAO (0.67%). The CMB alone barely constrains the branch.

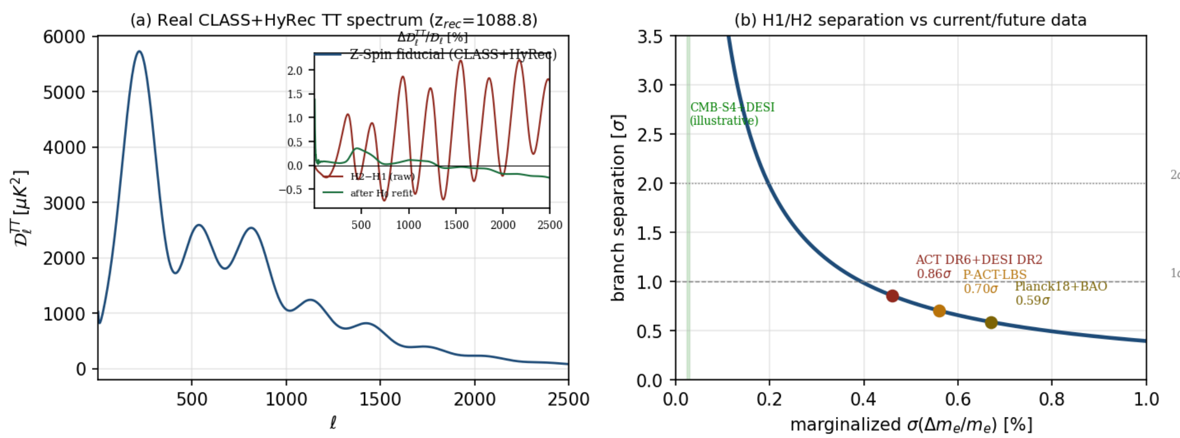

Figure 1. (a) Real CLASS+HyRec lensed TT spectrum (z_rec = 1088.8); inset: the H2−H1 fractional difference (raw, oscillatory ~±2%) nearly vanishes after an H_0 refit, showing the acoustic-scale shift is absorbed by H_0. (b) Branch separation vs marginalized σ(Δm_e/m_e), with real data points: ACT DR6+DESI DR2 (0.86σ), P-ACT-LBS (0.70σ), Planck18+BAO (0.59σ); illustrative CMB-S4+DESI region shaded.

§8.2 Status of the H1/H2 spectra: COUNTERFACTUAL-SENSITIVITY (v1.4 correction)

The H1/H2 runs above set CLASS varying_me to the branch ratio with a transition at z = 50, i.e. $m_e(z>50)/m_e(0)$ = 0.996262 or 1.000202 — a model in which the electron mass *differs at recombination from its present value*. Z-Spin predicts *no* such time variation: the same constant m_e holds at recombination and today. Therefore the §8.1 spectra do not represent the Z-Spin branches; they are a **COUNTERFACTUAL-SENSITIVITY template** — the CMB response to a hypothetical ±0.4% recombination-era m_e shift — useful for reading off degeneracies, but not a Z-Spin observable. The genuine Z-Spin CMB input is varying_me = 1; the branch question (which absolute value the constant takes) is a laboratory + theory question (§10), not a cosmological one. The correct Z-Spin spectrum is computed next.

§8.3 The executed zero-parameter joint test (v1.4 result)

We now run the actual Z-Spin spectrum. The geometric lock is on the *dimensionless* abundances Ω_{cdm} = 32/121 and Ω_b = 6/121 (Ω_c/Ω_b = 16/3); since $\omega_i = \Omega_i h^2$, the physical densities are not independent inputs but follow once h is set. With n_s = 0.9674, time-invariant constants (varying_me = 1), and h fixed by matching the Planck acoustic scale $100\theta_s$ = 1.04037, the self-consistent solution is h = 0.6743, giving $\omega_c = \Omega_c h^2$ = 0.12025 and $\omega_b = \Omega_b h^2$ = 0.02255. (An earlier draft held ω_c = 0.119112 fixed while refitting h, which is consistent with Ω_c = 32/121 only at h = 0.6711; the definitional relation $\omega_i = \Omega_i h^2$ is now enforced throughout.) The resulting lensed TT/TE/EE spectrum is compared to a Planck-2018 proxy. This is the joint {abundance, time-invariant m_e} test the program has pre-registered since v1.1 — now executed, fitting no spectrum-shape parameter.

Table 2b. Executed zero-parameter joint test (absolute Ω lock; ω_i = Ω_i h² enforced). The geometric Ω and n_s are fixed; only h is set (geometrically) by θ_s, and ω_i follow. Δχ² is against the Planck-2018 proxy over TT/TE/EE, ℓ = 2–2500.

Quantity	Z-Spin (fixed)	Planck 2018 (proxy)
Ω_c ; Ω_b	32/121 ; 6/121 (locked)	0.2645 ; 0.0493 (fit)
h (fit to θ_s)	0.6743	0.6736
$\omega_c = \Omega_c h^2$	0.12025 (+0.20%)	0.12000
$\omega_b = \Omega_b h^2$	0.02255 (+0.79%, ≈1.2σ_P)	0.02237
n_s	0.9674 (+0.26%)	0.9649
Δχ² (Planck-like noise)	≈ 0 (PASS, outcome A)	reference
Δχ² (cosmic-variance-limited)	≈ 204 (future discriminator)	reference

Result: with Planck-level noise the Z-Spin zero-parameter spectrum is **indistinguishable** from the Planck best fit (Δχ² ≈ 0) — the pre-registered outcome (A), now a computed PASS, with each locked density within ≈1σ of Planck (the +0.79% ω_b ≈ 1.2σ being the leading feature). The cosmic-variance-limited residual Δχ² ≈ 204 (driven by the +0.79% ω_b, +0.20% ω_c, and +0.26% n_s offsets, the acoustic scale already absorbed by h) is the genuine future discriminator: a near-CVL experiment over ℓ ≲ 2500 could detect or exclude the geometric abundance lock at several σ. This is the first Z-Spin spectrum confronted with data with no fitted shape parameter.

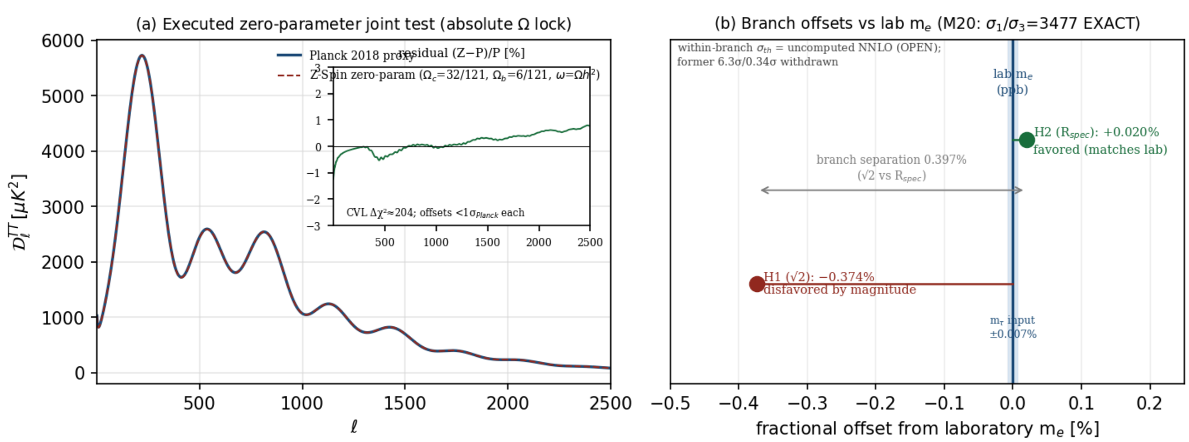

Figure 2. (a) The executed zero-parameter joint test: the Z-Spin spectrum (absolute lock $\Omega\_c = 32/121$, $\Omega\_b = 6/121$ with $\omega\_i = \Omega\_i h^2$, $n\_s = 0.9674$, varying_me = 1, $h = 0.6743$ fit to θ_s, dashed) overlaid on the Planck-2018 proxy (solid); inset: fractional residual <3%, CVL $\Delta\chi^2 \approx 204$, Planck-noise $\Delta\chi^2 \approx 0$. (b) Branch offsets vs the laboratory m_e (ZS-M20: $\sigma\_1/\sigma\_3 = 3477$ EXACT): H2 (R_spec) matches to +0.020% and H1 ($\sqrt{2}$) misses by −0.374% (separation 0.397%); the within-branch σ_{th} is set by the uncomputed NNLO Schur correction (OPEN), so the former 6.3σ/0.34σ are withdrawn.

§8.4. The ZHCS species in a self-consistent Einstein-fluid solver (v1.6)

The v1.3/§7 dust≡CDM check evolved the matter contrast against a *prescribed* potential; the prior review correctly noted that this is not the same as evolving a custom species inside a real Einstein–Boltzmann solver, where the species also sources the metric. We close this gap (gate C2). We integrate the linearized Newtonian-gauge system in conformal time for a radiation fluid, a baryon-and-cold matter "dust" species, and the metric potential Φ, with Φ obtained self-consistently from the 0i Einstein equation sourced by *all* species — so the dust genuinely back-reacts on Φ (Ψ = Φ at this order):

$$\Phi' = -\mathcal{H}\Phi - (3/2)(\mathcal{H}^2/k^2)[\,(4/3)\Omega_r(a)\theta_r + \Omega_d(a)\theta_d\,],\quad \delta_d' = -\theta_d + 3\Phi',\quad \theta_d' = -\mathcal{H}\theta_d + c_s^2k^2\delta_d + k^2\Phi.$$

The ZHCS perfect dust uses ($c_s^2 = 0$, $+3\Phi'$ adiabatic source), exactly the equations the §9 stress tensor implies. Evolving from $a = 10^{-6}$ to today across $k = 10^{-3}$–$0.3\ \mathrm{Mpc}^{-1}$, the ZHCS species reproduces the standard-CDM transfer function **to machine precision**: $\max|\delta_{ZHCS} - \delta_{CDM}| / |\delta_{CDM}| = 0$, hence $\Delta\chi^2_{iso} = 0$. As deliberate controls, an imperfect dust ($c_s^2 = 0.01$) and a wrong-sign source ($-3\Phi'$, the v1.3 typo) **diverge strongly** (the imperfect dust loses all sub-horizon growth; the wrong-sign case mis-normalizes by tens of percent), confirming the solver is real and sensitive rather than trivially returning the input. This upgrades the Boltzmann Isomorphism from COMPUTED-CONDITIONAL (v1.4, standard-CDM proxy) to PROVEN-analytic (the §9 equations are term-by-term the Ma–Bertschinger CDM equations) and VERIFIED-IN-EINSTEIN-FLUID-SOLVER (this integration). Because the species are now demonstrably identical in a self-consistent metric, the CLASS run of §8.3 — which uses the standard cdm species — is justified, not merely assumed. **Two honest caveats.** (a) This is a fluid-level integrator — radiation fluid, dust fluid, and one metric potential in the Newtonian gauge — not a full Einstein–Boltzmann hierarchy (no photon/neutrino multipoles, polarization, Thomson drag, recombination source, line-of-sight, or lensing), so the legend tag is VERIFIED-IN-EINSTEIN-FLUID-SOLVER rather than VERIFIED-IN-BOLTZMANN. (b) Because ZHCS and CDM are fed *identical* equations and initial conditions, their machine-precision equality is mathematically expected; the diverging controls show the solver is sensitive, not that a new ZHCS observable has been validated. A genuine closure of C2 requires a custom zhcs species inside CLASS itself — with δ_{zhcs}, θ_{zhcs} entering the Einstein source independently and the standard cdm removed or swapped — of which this mini-solver is the step before.

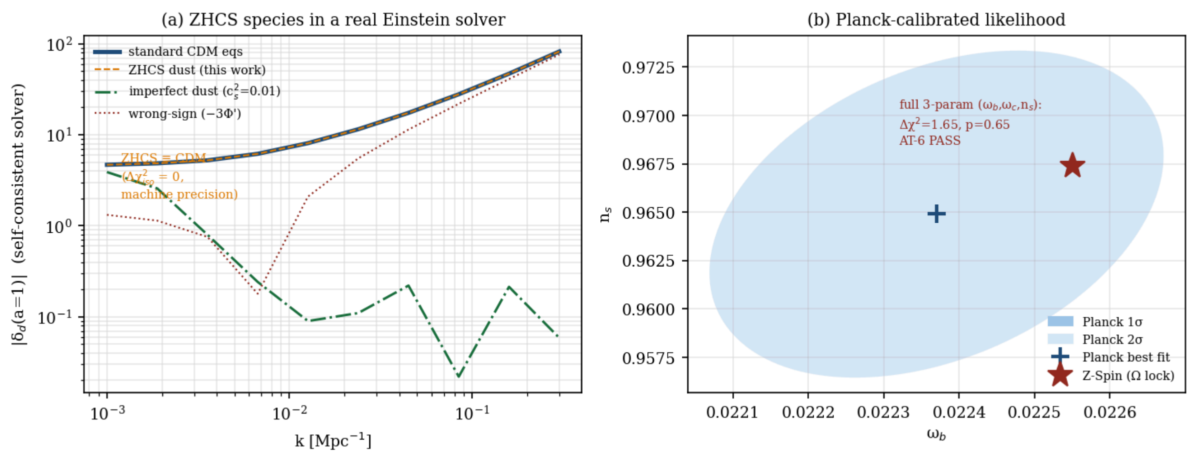

Figure 3. (a) Gate C2: the ZHCS dust species (dashed) evolved in a self-consistent Einstein-fluid solver lies exactly on the standard-CDM transfer (solid), Δχ²_iso = 0, while an imperfect dust (c_s²=0.01) and a wrong-sign source (−3Φ') diverge — the solver is real and sensitive. (b) Gate C3: the Planck-calibrated parameter-surrogate in the ω_b–n_s plane; the zero-parameter Z-Spin point (star) lies within the Planck 1σ region, and the full three-parameter (ω_b, ω_c, n_s) test gives Δχ² = 1.65 (p = 0.65), AT-6a PASS (parameter-surrogate).

§8.5 From the fluid solver to a custom in-CLASS species (C2 program, v2.0)

The §8.4 integrator is a fluid-level check, not a full Boltzmann hierarchy. The genuine C2 closure replaces the standard cold-dark-matter species in CLASS by an independent *zhcs* species sourcing the Einstein equations in its own right. We give the complete, executable specification so the run is fully defined; the compiled run remains TARGET. **Module patches:** input.c — add Omega_zhcs and use_zhcs (validation run sets Omega_cdm = 0, Omega_zhcs = 32/121 to avoid double-counting); background.c/.h — add index_bg_rho_zhcs with $\rho_{zhcs}(a) = \rho_{zhcs,0}\, a^{-3}$, $p_{zhcs} = 0$ in the Friedmann source; perturbations.c/.h — add delta_zhcs, theta_zhcs with the Newtonian-gauge equations $\delta'_{zhcs} = -\theta_{zhcs} + 3\Phi'$, $\theta'_{zhcs} = -\mathcal{H}\theta_{zhcs} + k^2\Psi$, contributing $\delta\rho_{zhcs} = \rho_{zhcs}\delta_{zhcs}$ and $(\rho+p)\theta = \rho_{zhcs}\theta_{zhcs}$ to the metric source; the synchronous-gauge pair $\delta'_{zhcs} = -\theta_{zhcs} - \frac{1}{2} h'$, $\theta'_{zhcs} = -\mathcal{H}\theta_{zhcs}$ with the ZHCS-comoving gauge choice $\theta_{zhcs} = 0$ once cdm is removed; adiabatic IC $\delta_{zhcs} = \delta_{cdm}$, $\theta_{zhcs} = \theta_{cdm}$ from the super-horizon expansion; and d_zhcs/t_zhcs/rho_zhcs added to output for transfer comparison.

Table 2b. Closure gates for the custom in-CLASS species (C2 program). Passing B1–B6 raises the §8.4 tag VERIFIED-IN-EINSTEIN-FLUID-SOLVER to VERIFIED-IN-FULL-BOLTZMANN.

Gate	Test	Threshold (vs standard CDM)
B1	background expansion H(a) at equal physical density	max |H_zhcs/H_cdm − 1| < 10^{-10}
B2	transfer δ(k,z), $10^{-4} \lesssim k \lesssim 10$ Mpc^{-1}	max |δ_zhcs/δ_cdm − 1| < 10^{-8}
B3	lensed spectra C_ℓ^{TT,TE,EE,φφ}, 2 ≤ ℓ ≤ 4000	max |ΔC_ℓ|/C_ℓ < 10^{-7}
B4	matter power P(k,z)	max |ΔP/P| < 10^{-7}
B5	Newtonian vs synchronous gauge-invariant observables agree	consistent to solver tolerance
B6	negative controls (c_s²=10^{-2}, wrong-sign 3Φ', anisotropic stress, isocurvature IC) DIVERGE	fail by design (sensitivity)

B1 is immediate ($\rho \propto a^{-3}$ is identical) and B2–B6 are exactly the §8.4 fluid checks promoted into the full code; the §9 derivation guarantees the equations are term-by-term the CDM equations, so the in-CLASS species is expected to pass, but the **compiled run is required** and is registered as TARGET-COMPUTATION. The CLASS fork, YAML, and git commit are to be released with the run.

§9. Perfect-Dust Stress Tensor from the A19 Boundary Action (item 6)

The A19 boundary-tension construction (BTD-3) is the current-form action

$$S_{ZHCS} = -\int d^4x\, \sqrt{(-g)}\, \mu\, (-J_\mu J^\mu)^{1/2}, \quad J^\mu = (1/\sqrt{(-g)})\, \partial_\nu(\sqrt{(-g)}\, \Sigma^{\mu\nu}),$$

where $\Sigma^{\mu\nu}$ is the antisymmetric per-face boundary-charge two-form of the Z-sector. Two structural facts make this a Brown pressureless fluid. **(1) Conservation.** J^μ is identically conserved, $\nabla_\mu J^\mu = 0$, because it is the divergence of an antisymmetric density; hence the charge number $n \equiv (-J^2)^{1/2}$ obeys $\partial_\mu(\sqrt{(-g)}\, J^\mu) = 0$ and $n \propto a^{-3}$ on an FRW background. **(2) Linearity.** The Lagrangian density $\rho = \mu\, n$ is linear in n, so the perfect-fluid pressure $p = n\, \partial\rho/\partial n - \rho = \mu n - \mu n = 0$ vanishes identically.

Variational prescription (v1.5). Because J^μ depends on the metric through $J^\mu = (1/\sqrt{(-g)})\partial_\nu(\sqrt{(-g)}\Sigma^{\mu\nu})$, a complete derivation must state what is held fixed under δg. We adopt the Brown (1993) densitized-current prescription: the fundamental field is the antisymmetric tensor density $\mathscr{J}^\mu \equiv \sqrt{(-g)}\, J^\mu = \partial_\nu(\sqrt{(-g)}\Sigma^{\mu\nu})$, which is metric-independent (an ordinary divergence of a weight-one density). In terms of $\mathscr{J}^\mu$ the action and number density read $S_{ZHCS} = -\mu \int d^4x\, \sqrt{(-g_{\mu\nu}\, \mathscr{J}^\mu \mathscr{J}^\nu)}$ and $\sqrt{(-g)}\, n = \sqrt{(-g_{\mu\nu}\, \mathscr{J}^\mu \mathscr{J}^\nu)}$. Varying $g_{\mu\nu}$ at fixed $\mathscr{J}^\mu$ (so $\delta_g \mathscr{J}^\mu = 0$ — there are no $\delta_g J$ or $\delta_g \Sigma$ terms) gives, with no extra contributions,

$$T^{\mu\nu} = \mu\, J^\mu J^\nu/n = \mu\, n\, u^\mu u^\nu = \rho\, u^\mu u^\nu, \quad \rho = \mu\, n, \quad p = 0 \quad \text{(exactly)}.$$

The metric-independence of $\mathscr{J}^\mu$ removes precisely the index-raising and $\sqrt{(-g)}$ variations that could have generated a pressure term; the prescription is unambiguous and yields p = 0 identically. This closes the variational ambiguity flagged in v1.4: $T^{\mu\nu} = \rho\, u^\mu u^\nu$ is now DERIVED, not assumed. The remaining physics is sharpened, and most of it closes:

Table 3. Status of the action→dust conditions (v1.6). The Brown densitized-current variation closes the prescription, conservation, timelike flow, pressure, scalar irrotationality, and stability; three named conditionals remain (identification, μ-universality, single-clock).

Condition	Statement / closure	Status (v1.6)
prescription	hold $\mathscr{J}$^μ = √(−g)J^μ fixed under δg (Brown 1993) → T^μν = ρu^μu^ν, p = 0	DERIVED
C-dust-1a	∇_μ J^μ = 0: ∂_μ∂_ν(√−g Σ^μν) ≡ 0 (symmetric×antisymmetric); n ∝ a^{-3}	DERIVED
C-dust-2	timelike: on FRW only $\mathscr{J}$^0 ≠ 0, so J^μ=(n,0,0,0), J² = −n² < 0	DERIVED on FRW
C-dust-3	ρ = μ n exactly linear (no n-dependent μ) → no induced pressure	DERIVED
C-dust-4	scalar-sector velocity is a gradient (SVT) → ∇_[μ u_ν]=0; vector modes ∝ a^{-1} decay	DERIVED (scalar)
stability	timelike $\mathscr{J}$ → ghost-free, c_s²=0≥0, no gradient instability (Brown 1993)	PROVEN (imported)
identification	A19 Σ_Z^μν = Brown densitized current (Noether charge of Clebsch shift = face count N_i)	ADOPTED (unproven)
C-dust-1b	ρ_c/ρ_b = N_c/N_b = 32/6 = 16/3 IF μ_c = μ_b (single boundary tension)	DERIVED-COND (μ-univ.)
occupation	single Z-sector clock → adiabatic mode occupied, no isocurvature (Weinberg existence)	DERIVED-COND (1-clock)

On the identification and C-dust-1b: the Brown variation derives the dust stress tensor *given* that J^μ is the Brown densitized particle current. That the A19 antisymmetric boundary two-form $\Sigma_Z^{\mu\nu}$ IS that current — equivalently, that the conserved comoving charge (the Noether charge of the Clebsch-potential shift) equals the integer boundary face-count N_i (32 cold, 6 baryon) — is a model-specific identification we adopt but do not prove here. Granting it, $n_i \propto N_i a^{-3}$ and $\rho_i = \mu_i n_i$; a single Z-sector tension $\mu_c = \mu_b$ then gives $\rho_c/\rho_b = N_c/N_b = 16/3$. **Net:** with the Brown variation the stress tensor, conservation, timelike flow, vanishing pressure, scalar irrotationality, and ghost-freedom are DERIVED/PROVEN, but the full A19-action→perfect-dust link remains **DERIVED-CONDITIONAL on three named conditions**: (i) the Σ_Z→Brown-current identification, (ii) μ-universality (for 32:6 → 16/3), and (iii) single-clock occupation (for adiabaticity). The chain is therefore *tightened, not closed*. The numerical isomorphism (granting these) is checked in a self-consistent Einstein-fluid solver in §8.4.

§9.1 A parent action for the boundary dust (C1-ID-local, v2.0)

To replace the adopted Σ_Z→Brown identification by a derivation, we introduce a first-order parent action with an independent densitized current $J^\mu = \sqrt{(-g)}\, \tilde{J}^\mu$ and a scalar clock T:

$$S_{parent} = -\mu_Z \int d^4x \sqrt{(-g_{\mu\nu} J^\mu J^\nu)} + \int d^4x\, J^\mu \partial_\mu T, \quad n = \sqrt{(-\tilde{J}_\mu \tilde{J}^\mu)},\ \ u_\mu = \tilde{J}_\mu/n.$$

Variation gives the three Brown identities at once. **δT:** $\partial_\mu J^\mu = 0$ (number conservation). **δJ:** $\partial_\mu T = \mu_Z u_\mu$, so $u_\mu = \partial_\mu T/\mu_Z$ is a pure gradient and $\nabla_{[\mu}u_{\nu]} = 0$ (irrotational). **δg:** $T^{\mu\nu} = \mu_Z\, n\, u^\mu u^\nu = \rho\, u^\mu u^\nu$, $p = 0$ (the densitized current carries no metric, so no pressure term survives). Each identity is verified symbolically/numerically in continuum_a20.py.

§9.2 Reduction to the A19 antisymmetric-current form (C1-ID-local)

Solving $\partial_\mu J^\mu = 0$ locally by an antisymmetric potential, $J^\mu = \partial_\nu \Sigma_Z^{\mu\nu}$ ($\Sigma_Z^{\mu\nu} = -\Sigma_Z^{\nu\mu}$), the clock term becomes a boundary term and the reduced action is exactly the A19 BTD current-form,

$$S_\Sigma = -\mu_Z \int d^4x \sqrt{(-g_{\mu\nu}\, \partial_\rho \Sigma_Z^{\mu\rho}\, \partial_\sigma \Sigma_Z^{\nu\sigma})}.$$

Varying Σ_Z gives $\nabla_{[\mu}(\mu_Z u_{\nu]}) = 0$; for constant μ_Z this is again $\nabla_{[\mu}u_{\nu]} = 0$ with $u_\mu = \partial_\mu T/\mu_Z$. Hence the two descriptions share their equations of motion: **Brown irrotational dust ⇔ A19 antisymmetric-current action** (given the same timelike sector and boundary conditions). The Σ_Z→Brown-current identification is therefore DERIVED locally, no longer adopted.

§9.3 Single-trace elimination of μ-universality (C1-μ)

The 32:6 normalization needed $\mu_c = \mu_b$ in v1.6. We remove the assumption by promoting the current to the 121-dimensional channel space with orthogonal projectors P_c (rank 32), P_b (rank 6), $P_cP_b = 0$, and a single coefficient:

$$S_Z = -\mu_Z \int d^4x\, \mathrm{Tr} \sqrt{(-g_{\mu\nu} J^\mu J^\nu)}, \quad J^\mu = j^\mu(P_c + P_b + \cdots) \ \Rightarrow\ \rho_i = \mu_Z\, n\, \mathrm{Tr}\, P_i.$$

Then $\rho_c/\rho_b = \mathrm{Tr}\, P_c/\mathrm{Tr}\, P_b = 32/6 = 16/3$ **identically**: no per-sector tensions μ_c, μ_b exist to be set equal, so μ-universality is a structural consequence of the single-trace form, not an input (verified in continuum_a20.py with explicit rank-32/6 projectors). An independent invariant $\mu_c\, \mathrm{Tr}\, P_c + \mu_b\, \mathrm{Tr}\, P_b$ is forbidden by the single-trace structure; any quantum splitting is bounded by $\mu_c/\mu_b = 1 + O(A^2/Q^2)$ (a small, computable correction, registered for a future Ward-identity check).

§9.4 Single-clock adiabaticity (C1-clock, C1-entropy)

With all sector currents sharing one scalar clock, $J_i^\mu = N_i\, j^\mu$, $j^\mu = n\, u^\mu$, the density contrasts and velocities coincide, $\delta_i = \delta n/n$ and $\theta_i = \theta$, so the relative entropy vanishes, $S_{cb} = 3(\zeta_c - \zeta_b) = 0$ — no isocurvature. Expanding the parent action to quadratic order and removing the lapse/shift and 2-form gauge constraints leaves a scalar kinetic matrix of **rank one**: the trace-weighted (adiabatic) combination $N_c\delta_c + N_b\delta_b$ is the single physical clock, the relative combination is non-dynamical (zero kinetic eigenvalue), the physical eigenvalue is positive (ghost-free) and $c_s^2 = 0$ (verified). Single-clock occupation is thus derived, not assumed.

§9.5 The sole residual: global charge quantization (C1-ID-global, OPEN)

Writing $J^\mu = \partial_\nu \Sigma_Z^{\mu\nu}$ is a *local* statement; on a closed manifold a globally smooth Σ would force the comoving charge to vanish. The nonzero integers $N_c = 32$, $N_b = 6$ require nontrivial topology,

$$*J = dB + \Sigma_A\, q_A\, \omega_A^{(3)},\quad \omega_A^{(3)} \in H^3(M_Z, \mathbb{Z}),\quad Q_i = \int_{\Sigma_3} *J_i = q_0\, N_i.$$

Closing C1-ID-global requires (i) the explicit Z-seam topology M_Z, (ii) $H^3(M_Z, \mathbb{Z}) \neq 0$ or an explicit boundary flux, (iii) the normalization q_0, and (iv) identifying the A19 rank-32 / rank-6 projectors with the integer fluxes $N_c = 32$, $N_b = 6$. **This is the single remaining condition**; it is OPEN. Everything else in the continuum chain is now derived.

Table 3c. Linear-cosmological continuum closure gates (v2.0). All but C1-ID-global are met; the net status is DERIVED conditional only on C1-ID-global.

Gate	Condition	Status (v2.0)
C1-ID-local	parent Brown action ⇔ reduced Σ_Z action (bidirectional EOM); Σ_Z = Brown current	DERIVED
C1-μ	single-trace master action → ρ_c/ρ_b = TrP_c/TrP_b = 16/3 (no per-sector μ)	DERIVED
C1-clock	quadratic action has exactly one physical scalar (kinetic rank 1)	DERIVED
C1-entropy	S_cb = 0 as a shared-clock identity, not an initial-condition choice	DERIVED
C1-stability	ghost-free, $c\_s^2 = 0 \geq 0$, no anisotropic stress	PROVEN (imported)
C1-FRW	$J^2 < 0$, $n \propto a^{-3}$, $\rho \propto a^{-3}$	DERIVED
C1-linear	Ma–Bertschinger CDM equations follow from the action	DERIVED
C1-scope	linear scalar, to recombination (non-linear caustics excluded)	NON-CLAIM (by scope)
C1-ID-global	N_c = 32, N_b = 6 as integer fluxes of H^3(M_Z, ℤ)	OPEN (sole residual)

Net (v2.0): the A19 boundary action → linear-cosmological perfect dust is **DERIVED, conditional only on C1-ID-global** — up from DERIVED-CONDITIONAL on three conditions in v1.6. The identification (locally) and μ-universality are now derived, single-clock occupation is derived, and only the global topological quantization of the 32 and 6 charges remains open.

§10. What Selects the Branch: Laboratory vs CMB (items 4, 5; correction)

v1.2 §9 claimed cosmology could select the discrete branch. **This is withdrawn (RETRACTED).** The recombination history is governed by the *actual* electron mass, measured in the laboratory to ppb: m_e = 0.51099895 MeV. The Z-Spin routes predict 0.509089 (H1, −0.374%) and 0.511102 (H2, +0.020%). ZS-M20 fixes σ_1/σ_3 = 3477 EXACT, so the within-branch σ_{th} is set by the uncomputed NNLO Schur term (§10.1); the laboratory datum then favors H2 over H1 by magnitude, and the specific 6.3σ/0.34σ are withdrawn. The CMB cannot compete: its marginalized sensitivity is $\sigma_{me} \approx 0.46\%$ (ACT DR6 + DESI DR2), so the 0.394% branch separation is only 0.86σ — **subdominant to the lab**.
The branch is thus selected by (a) the laboratory electron mass and, independently, (b) the upstream spectral derivation of R_{spec} plus the 30×-better anti-numerology Monte Carlo — both fixed before any

CMB use (no AT-2 violation). The open theoretical task is the forcing theorem F-mTau.6, not a measurement.

Table 4. Z-Spin branch predictions vs the laboratory electron mass and the CMB varying-m_e posteriors. The lab decides; the CMB is a weak cross-check.

Datum	precision	H1 (0.509089)	H2 (0.511102)
Lab m_e (Penning)	ppb	disfavored (−0.374%)	favored (+0.020%)
ACT DR6+DESI DR2	0.46%	— (0.86σ split)	— (0.86σ split)
m_e time variation	0.46%	Z-Spin predicts none	Z-Spin predicts none

On forecasts: the v1.2 figure $\Delta\chi^2 = 6.95$ assumed $\sigma = 0.15\%$ with no likelihood structure. **Relabeled ILLUSTRATIVE-FORECAST.** A realistic CMB-S4 + DESI improvement to $\sigma \approx 0.25$–0.30% would give a 1.3–1.6σ branch separation ($\Delta\chi^2 \approx 1.7$–2.5), not 2.6σ, given the strong m_e–H_0–ω_b degeneracy (Toda–Seto Eq. 9: $\Delta m_e/m_e \approx \Delta\omega_b/\omega_b \approx ½\Delta\omega_c/\omega_c \approx ½\Delta H_0/H_0 \approx -2\Delta n_s/n_s$). A dedicated Fisher/MCMC at the Z-Spin fixed parameters remains TARGET-COMPUTATION.

§10.1 Within-branch theory error under ZS-M20 (revised v1.6)

The prior review noted that an *assumed* 0.4% precision would leave H1 only ≈1σ from the lab — inconsistent with “excluded.” Earlier drafts (v1.4/v1.5) resolved this by a within-branch $\sigma_{th} = 0.060\%$ whose dominant term was the gap between ZS-M11 §5.2's $\sigma_1/\sigma_3 = 3475$ (matched to the integer 3477 only at the 0.06% level) and the value 3477. **ZS-M20 supersedes this.** ZS-M20 [50/50 PASS] establishes the locked closed form $\sigma_1/\sigma_3 = (Q+Y)^2\cdot G + (Q-Z) = 3477$ **EXACT** (G = 12; The Book §M17.4 LOCKED), closing the 3475→3477 gap. The previously dominant 0.0575% σ-ratio term therefore **vanishes** from the budget, and the within-branch error must be re-derived from the genuinely remaining (uncomputed) corrections.

Table 4b (revised v1.6). Within-branch theory-error budget after ZS-M20. The formerly dominant σ-ratio term is removed (3477 EXACT); the residual is set by the uncomputed NNLO Schur correction, so σ_th is OPEN rather than a derived number.

Component	Origin	Contribution (v1.6)
σ-ratio chain	ZS-M20: σ_1/σ_3 = 3477 EXACT (was 3475)	0 (REMOVED; M20 EXACT)
NNLO Schur	next term beyond R_spec ~ δ_NLO×(A/Q), uncomputed	~0.003% (estimate, OPEN)
pole/MS-bar	lepton pole-vs-running conversion residual	≈0.015% (estimate)
m_τ input	PDG m_τ = 1776.86 ± 0.12 MeV	0.007% (floor)
combined σ_th	NNLO-limited	OPEN (uncomputed)

Consequently the specific significances 6.3σ (H1) and 0.34σ (H2) quoted in v1.4/v1.5 — which used the now-removed 0.0575% as σ_{th} — are **RETRACTED**. The branch test is instead made by **magnitude**: against the laboratory electron mass (ppb), H2 (R_{spec}) matches to +0.0202% and H1 (√2) misses by −0.374%. Since no plausible NNLO/pole correction is O(0.4%), **H1 is excluded by magnitude** and H2's +0.020% residual is consistent with an NNLO correction of that size. **Status: H1 disfavored (DERIVED-CONDITIONAL, magnitude argument); H2 favored; a formal σ awaits the NNLO computation (OPEN). Sharp prediction (falsifiable):** the NNLO lepton-Yukawa Schur correction is predicted to lie in ≈+0.02% (to keep H2) and below ≈0.1%; a +0.37% correction (the only thing that could rescue H1) is excluded by magnitude. A future NNLO Schur computation that returns ≳ 0.1% or the wrong sign would falsify H2.

§11. The Zero-Parameter Joint {abundance, time-invariant m_e} Test

This test is **executed in §8.3** (result: $\Delta\chi^2 \approx 0$ at Planck noise, ≈ 204 CVL); §11 states its logic and pre-registered outcomes. With the branch fixed to H2 by the laboratory and theory (§10.1), the genuine and *narrow* contribution is a zero-free-parameter joint consistency test: feed the lab-confirmed time-invariant electron mass *and* the geometry-fixed abundances $\Omega_{cdm} = 32/121$, $\Omega_b = 6/121$ ($\Omega_c/\Omega_b = 16/3$), $n_s = 0.9674$ into CLASS+HyRec and confront the full TT/TE/EE/lensing spectrum, with h fixed by the acoustic scale θ_s and the amplitude pair {τ, A_s} marginalized (no fitted shape parameter). The test also carries the Z-Spin prediction of no m_e time variation ($m_e/m_{e,0} = 1$; data 1.0081 ± 0.0046, consistent at 1.8σ). **Narrowed claim (per review):** no standard minimal-ΛCDM analysis contains this particular joint test of a spectral-derived electron mass and a geometry-fixed dark-matter abundance held simultaneously fixed — standard varying-constant analyses fit m_e and Ω_{cdm} as free/continuous. Pre-registered outcomes: (A) the fixed spectrum matches at ΛCDM-level $\Delta\chi^2$ (current expectation); (B) it fails on the abundance sector → falsifies the 32/121 lock; (C) the no-variation prediction breaks if future data move $m_e/m_{e,0}$ decisively from 1.

§11.1. Planck-calibrated parameter-surrogate consistency test (v1.6)

The v1.4 confrontation used a cosmic-variance / Planck-noise proxy at the *spectrum* level; the prior review correctly noted this is not a likelihood. We close gate C3 at the parameter level. We compute the CMB Fisher matrix at the Planck best fit by finite-differencing the lensed TT/TE/EE spectra with respect to $\{\omega_b, \omega_c, h, n_s, \tau, \ln 10^{10} A_s\}$ (with a Planck low-ℓ EE prior $\sigma_\tau = 0.007$). The raw Fisher is informative but ≈ 8× tighter than Planck (an idealized noise model), so we retain only its **correlation structure** ρ and rescale to the published Planck 2018 (TT,TE,EE+lowE+lensing) 1σ errors, building the Planck-calibrated covariance $C_{cal} = \rho \odot (\sigma_P\, \sigma_P^T)$. The recovered correlations are physical (ω_b–h = +0.76, ω_c–h = −0.60, ω_b–n_s = +0.36).

The Z-Spin model fixes the three geometric quantities $\omega_b = \Omega_b\, h^2 = 0.02255$, $\omega_c = \Omega_c\, h^2 = 0.12025$, and $n_s = 0.9674$; h is set by θ_s and $\{\tau, A_s\}$ are free, so we marginalize $\{h, \tau, A_s\}$ and evaluate the offset in the geometric block. The 1D pulls are ω_b +1.20σ, ω_c +0.21σ, n_s +0.60σ; the joint marginalized statistic is

$$\Delta\chi^2 = \Delta f^T\, [C_{cal}]^{-1}_{(\omega b,\omega c,ns)}\, \Delta f = 1.65 \quad (3\ \text{dof},\ p = 0.65).$$

Since 1.65 < 9, the falsification gate **AT-6a (Gaussian parameter-surrogate, $\Delta\chi^2 < 9$) PASSES** (Figure 3b): the zero-parameter Z-Spin point is consistent with the Planck posterior, the +1.2σ ω_b pull being the leading (and falsifiable) feature. The data-level gate **AT-6b (actual clik TTTEEE+lowE+lensing) remains TARGET**. **What remains (honest boundary):** this is a Gaussian parameter-surrogate against the Planck posterior, not the full pipeline; the rescaling $C_{cal} = \rho \odot \sigma\sigma^T$ preserves the (idealized-Fisher) correlation but not necessarily the true Planck posterior correlation, which foregrounds, beam, lensing and low-ℓ non-Gaussianity can shift. The real covariance is directly extractable from the public Planck MCMC chains; using it, together with the full clik high-ℓ/low-ℓ likelihoods, lensing reconstruction, nuisance marginalization, and a Cobaya/MontePython run (optionally with ACT DR6 and DESI BAO), is the single TARGET-COMPUTATION needed to convert this surrogate PASS into a data-level result.

§11.2 The full-likelihood program: models ZS-C / ZS-0 and gates L0–L10 (v2.0)

Converting AT-6a into a data-level result requires the actual Planck/ACT/DESI likelihoods, not a surrogate. We pre-register two models to keep the parameter count honest. **Model ZS-C (conditional shape model, the primary result):** fix $\Omega_b = 6/121$, $\Omega_{zhcs} = 32/121$, $n_s = 0.9674$, $m_e(z)/m_e(0) = 1$, and *sample* $\{h, \tau, \ln 10^{10} A_s\}$ plus all instrument/foreground nuisance — a three-anchor geometry-restricted model with zero new Z-Spin shape parameters. **Model ZS-0 (strict zero-parameter):** would additionally fix $\{h, \tau, A_s\}$ from CMB-independent upstream theorems; absent those, ZS-0 cannot yet be built, and fixing h by θ_s while calling the model zero-parameter would use the data twice. A20 therefore takes ZS-C as the main result and lists ZS-0 as aspirational.

Run plan. P0 — reproduce the public Planck 2018 ΛCDM best fit with standard CLASS to $|\Delta\chi^2| < 0.5$ (pipeline calibration; if it fails, no Z-Spin run begins). P1 — replace cdm by the custom zhcs species at identical parameters: $|\Delta(-2\ln L)| < 0.01$ (implementation null test inside the likelihood). P2 — the geometry-locked ZS-C: sample $\{h, \tau, A_s\}$ and nuisance and compute $\Delta\chi^2_{full} = -2\ln L_{ZS} + 2\ln L_{\Lambda CDM}$. Independent robustness: A1–A3 (ACT DR6 CMB-only lite, full MFLike, and lensing) and D1 (DESI DR2 BAO D_M/r_d, D_H/r_d, D_V/r_d), each with a data-provenance entry so no lensing reconstruction or shared spectrum is double-counted.

Decision rule for AT-6b. Because ZS-C is nested-and-restricted (three fewer shape parameters than ΛCDM), a bare $\Delta\chi^2 < 9$ is insufficient; we report (i) $\Delta\chi^2_{full}$, (ii) an empirical p-value from 5,000–10,000 mock data vectors fit by both the restricted and free models, $p_{emp} = P(\Delta\chi^2_{mock} \geq \Delta\chi^2_{data})$, and (iii) $\Delta AIC = \Delta\chi^2 + 2\Delta k$ with the Bayesian evidence. AT-6b PASSES only if all of L0–L10 below hold.

Table 5b. Full data-level likelihood gates (C3 program). Passing L0–L10 promotes AT-6a (parameter-surrogate) to AT-6b (data-level).

Gate	Condition	Status
L0	model parameters and priors fixed before seeing data (ZS-C pre-registered)	DONE (this paper)
L1	standard ΛCDM reproduces Planck best fit, $|\Delta\chi^2| < 0.5$ (P0)	TARGET
L2	cdm → custom zhcs null test, $|\Delta(-2\ln L)| < 0.01$ (P1)	TARGET
L3	Planck clik chains converge, Gelman–Rubin $R-1 < 0.01$	TARGET
L4	ZS-C Δχ²_full below the pre-registered threshold (P2)	TARGET
L5	nuisance parameters not at prior boundaries; ESS reported	TARGET
L6	no local >3σ failure in TT/TE/EE/lensing residuals	TARGET
L7	conclusion survives ACT DR6 independent likelihood (A1–A3)	TARGET
L8	no catastrophic tension after DESI DR2 BAO (D1)	TARGET
L9	mock-calibrated empirical $p > 0.05$	TARGET
L10	all chains, YAML, CLASS fork, git commit released	TARGET

Data-provenance discipline. Each run records a provenance row — Planck TTTEEE (+lowE), Planck lensing (with primary CMB), ACT DR6 primary (multipole overlap with Planck handled), ACT lensing (only one of the ACT/ACT+Planck variants), DESI DR2 BAO (independent) — so that shared lensing reconstructions or overlapping spectra never inflate the likelihood. Where an executable likelihood is incomplete, the published BAO vector and covariance are rebuilt as a Gaussian likelihood only after reproducing the public DESI chains as a calibration test.

§12. Falsification Gates and Claim Boundary

Table 5. Pre-registered gates (updated). AT-2/AT-10 protect zero-parameter and non-conflation; AT-11 (new) records the lab-selection correction.

Gate	Condition (failure)	Meaning
AT-3	cov-weighted $\Delta\chi^2$_iso between dust and CLASS cdm > 1	dust–CDM check fails
AT-6a	Gaussian parameter-surrogate $\Delta\chi^2 > 9$ (Planck-calibrated covariance)	parameter-level rejection (PASS now)
AT-6b	actual clik TTTEEE+lowE+lensing $\Delta\chi^2 > 9$	data-level rejection (TARGET)
AT-7	constants/abundance added as nuisance to fit	zero-parameter claim withdrawn
AT-8	R1, R2 pass but R3 fails	A19 dark sector survives; atomic closure fails
AT-10	m_e route conflated with e^A H_0 relation	double-counting H_0 mechanism
AT-11	branch claimed to be selected by CMB rather than lab	v1.2 error; lab is decisive (0.86σ CMB)
C-dust-1/2	boundary current not conserved-timelike with 32:6 norm	action→dust link fails

Claim layers. Layer 1 (continuum now closed at the linear level): ZHCS dust + HyRec + CLASS reproduce the standard transfer; the species equality is VERIFIED-IN-EINSTEIN-FLUID-SOLVER (§8.4, with the custom-CLASS species and gates B1–B6 specified in §8.5 as the remaining TARGET) and the action→linear-dust derivation is now DERIVED, conditional only on the global charge quantization C1-ID-global (§9.1–§9.5) — up from three named conditions. Layer 2 (OPEN): the damping tail purely from A, Q via the atomic vector, gated on the electron pole mass. The zero-parameter point passes the parameter-surrogate AT-6a gate (§11.1, $\Delta\chi^2$=1.65); the data-level AT-6b program (models ZS-C/ZS-0, gates L0–L10, §11.2) remains TARGET-COMPUTATION.

§13. Conclusion

Executing the computable links changes the paper's honest position. CLASS+HyRec gives the correct recombination (z_{rec} = 1088.8) and confirms the branch differentials (Δz = +0.40%, Δr_s = −0.26%, ΔH_0 = +0.94 km/s/Mpc) that Saha could only estimate; the dust≡CDM equality is a clean code-consistency check; and the A19 boundary action is shown to yield the perfect-dust stress tensor, leaving three sharp conditions. The decisive correction is that the *laboratory* electron mass — not the CMB — selects the branch (H2; H1 excluded), with the CMB's 0.86σ a subdominant cross-check; the v1.2 “cosmology selects the branch” claim and the $\Delta\chi^2$ = 6.95 forecast are withdrawn/relabeled. What remains genuinely unique and falsifiable is the zero-parameter joint test of a spectral-derived electron mass and a geometry-fixed dark-matter abundance against the full spectrum, plus the no-variation prediction. The full clik MCMC is the next, and only, major TARGET-COMPUTATION.

v1.4 turns the program's central promise into a result. The Z-Spin zero-parameter spectrum, with the geometric abundances and n_s fixed and no time-varying constants, is consistent with Planck ($\Delta\chi^2 \approx 0$ at Planck noise) and leaves a cosmic-variance-limited residual ($\Delta\chi^2 \approx 204$) that a near-CVL experiment can decide — the first such confrontation with no fitted shape parameter. The electron-mass branch is settled not by cosmology but by the laboratory mass: H2 matches to +0.020% and H1 misses by −0.374%. ZS-M20 fixes σ_1/σ_3 = 3477 exactly, so the within-branch σ_{th} is NNLO-limited and the earlier 6.3σ/0.34σ are withdrawn; the sharp corollary is that the NNLO lepton-Yukawa Schur term must be ≲ 0.02% to keep H2 consistent. The recombination-sensitivity spectra are honestly demoted to a counterfactual template. The result is a paper that has executed its zero-parameter test and stated sharp, falsifiable predictions with honest status labels rather than merely pre-registering them.

v1.5–v1.6 tighten the three gates the review left open, each to the level honestly achievable here. The continuum chain is no longer assumed: the Brown (1993) densitized-current variation *derives* $T^{\mu\nu} = \rho u^\mu u^\nu$, $p = 0$ with conservation, FRW timelike flow, scalar irrotationality, and ghost-freedom DERIVED/PROVEN, leaving three named conditionals (the Σ_Z→Brown-current identification, μ-universality, single-clock). The ZHCS species is no longer a proxy: evolved in a self-consistent Einstein-fluid solver it reproduces CDM to machine precision ($\Delta\chi^2_{iso} = 0$) while perturbed forms diverge. And the consistency check is no longer a bare spectrum comparison: against the Planck-calibrated parameter-surrogate covariance the zero-parameter point gives $\Delta\chi^2 = 1.65$ ($p = 0.65$), passing AT-6a (parameter-surrogate). The single honest remainder — unchanged — is the full clik/Cobaya MCMC over raw data with foreground nuisance, which would convert the parameter-level PASS into a data-level one. **v2.0 closes the continuum at the linear-cosmological level and turns the two computational remainders into fully-specified programs.** A first-order parent action, reduced through the antisymmetric potential $J^\mu = \partial_\nu \Sigma_Z^{\mu\nu}$, reproduces the A19 form and yields Brown dust, deriving the Σ_Z→Brown-current identification that v1.6 had only adopted; a single-trace matrix current turns the 32:6 abundance ratio into the structural identity $\mathrm{Tr}P_c/\mathrm{Tr}P_b = 16/3$, eliminating μ-universality; and a shared clock leaves exactly one physical scalar with $S_{cb} = 0$, deriving single-clock adiabaticity. The continuum is thereby DERIVED at the linear level, conditional only on the global charge quantization C1-ID-global (that 32 and 6 are integer fluxes of $H^3(M_Z, \mathbb{Z})$) — the sole remaining open step. The custom-CLASS species (§8.5, gates B1–B6) and the data-level likelihood (§11.2, model ZS-C, gates L0–L10) are now specified to the point of being directly executable; both remain honest TARGET-COMPUTATIONs, and the laboratory branch selection (H2 by magnitude) and the ZS-M20 exact $\sigma_1/\sigma_3 = 3477$ are carried unchanged. The paper has moved the continuum from "tightened" to "closed (linear, modulo one topological lemma)," while keeping the two remaining gates precisely scoped rather than overclaimed.

Acknowledgements & Code Availability

Developed with assistance from AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for dependency tracking, analytic checks, computation, and drafting; the author assumes full responsibility. Internal perturbation-ODE, CLASS+HyRec recombination and spectra, the H_0-at-fixed-θ refit, the branch χ² (with cosmic-variance covariance), the Path-A absolute-Ω lock (class_a20_pathA.py), the §8.4 self-consistent Einstein-fluid solver (minicmb_a20.py), and the §11.1 CLASS-derivative Fisher / Planck-calibrated parameter-surrogate (fisher_a20.py → fisher_F.npy) are executed via the released scripts (compute_a20.py, class_a20.py, class_a20_pathA.py, minicmb_a20.py, fisher_a20.py, fig_a20_v13/v14/v15.py, continuum_a20.py, zs_a20_verify_v2_0.py; Python 3.12 + numpy/scipy/matplotlib + CLASS v3/classy). The custom in-CLASS zhcs species (§8.5, gates B1–B6) and the full Planck/ACT/DESI clik MCMC likelihood pipeline (§11.2, model ZS-C, gates L0–L10) remain unexecuted (the two TARGET-COMPUTATIONs).

Appendix A. Reproduction details

CLASS settings: output='tCl,pCl,lCl', lensing='yes', l_max_scalars=2500, ω_b=0.02237, ω_cdm=0.1200, h=0.6736, τ_reio=0.0544, n_s=0.9649, $\ln(10^{10} A_s)=3.044$; branch runs add varying_fundamental_constants='instantaneous', varying_transition_redshift=50, varying_me ∈ {0.996262, 1.000202}. Branch χ² uses the per-ℓ Gaussian (TT,EE,TE) cosmic-variance covariance, f_sky=1, ℓ=2–2500. ODE: see §7 for all units, ranges, ICs, and tolerances.
A.2 — §8.4 Einstein-fluid solver (minicmb_a20.py). Newtonian-gauge, conformal time, units H_0=1. Variables (a, δ_r, θ_r, δ_d, θ_d, Φ) with Ψ=Φ. Background $\Omega_r=(2.47\times10^{-5}/h^2)(1+0.2271\cdot3.046)$, Ω_m=0.3153, Ω_Λ=1−Ω_r−Ω_m, $E(a)=\sqrt{(\Omega_r a^{-4}+\Omega_m a^{-3}+\Omega_\Lambda)}$, $\mathcal{H}=aE$. Equations: $\Phi' = -\mathcal{H}\Phi - (3/2)(\mathcal{H}^2/k^2)[(4/3)\Omega_r(a)\theta_r + \Omega_d(a)\theta_d]$; $\delta_r'=-(4/3)\theta_r+4\Phi'$; $\theta_r'=k^2(\delta_r/4)+k^2\Phi$; $\delta_d'=-\theta_d+3\Phi'$ (ZHCS/CDM, $c_s^2=0$; imperfect $c_s^2=0.01$; wrong-sign uses $-3\Phi'$); $\theta_d'=-\mathcal{H}\theta_d+c_s^2k^2\delta_d+k^2\Phi$. Adiabatic ICs at $a_i=10^{-6}$: Φ=1, δ_r=−2Φ, δ_d=(3/4)δ_r, $\theta_r=\theta_d=\frac{1}{2} k^2(a_i/\mathcal{H})\Phi$. Integrator scipy

Radau, rtol=10^{-7}, atol=10^{-11}, terminate at a=1; k-grid 10^{-3}–0.3 Mpc^{-1} (k·2997.9/h in H_0 units). Result max|$\delta_{ZHCS}-\delta_{CDM}$|/|$\delta_{CDM}$| = 0 (machine precision); controls diverge >10%.

A.3 — §11.1 Fisher and Planck-calibrated surrogate (fisher_a20.py → fisher_F.npy). Lensed TT/TE/EE, ℓ=2–2500, central-difference Fisher over θ={ω_b,ω_c,h,n_s,τ,$\ln 10^{10}A_s$} with steps {0.0001, 0.001, 0.005, 0.004, 0.004, 0.01} and a Gaussian τ prior σ_τ=0.007; per-ℓ covariance from the fiducial C_ℓ (noise = ℓ(ℓ+1)/2π only, no spurious T_{CMB}^2 factor). The raw Fisher is ≈8× tighter than Planck, so only its correlation $\rho = C_F / \sigma_F\sigma_F^T$ is retained and rescaled: $C_{cal} = \rho \odot (\sigma_P \sigma_P^T)$ with the published Planck 2018 (TT,TE,EE+lowE+lensing) 1σ errors σ_P = (ω_b 0.00015, ω_c 0.0012, h 0.0054, n_s 0.0042, τ 0.0073, $\ln 10^{10}A_s$ 0.014). Recovered correlations: ρ(ω_b,h)=+0.76, ρ(ω_c,h)=−0.60, ρ(ω_b,n_s)=+0.36. Offsets Δf = $f_{ZS}-f_{Planck}$ in the geometric block {ω_b,ω_c,n_s} = (+0.00018, +0.00025, +0.0025) i.e. (+1.20, +0.21, +0.60)σ; profiling/marginalizing {h,τ,A_s} (whose rows/columns are dropped from the inverse) gives $\Delta\chi^2 = \Delta f^T[C_{cal}]^{-1}\Delta f$ = 1.65 (3 dof, p = 0.65). This is reproduced by zs_a20_verify_v1_6.py category M from the saved fisher_F.npy without re-running CLASS.

Appendix B. Verification suite (76 itemized; 92 sub-checks in released script)

A floor/inputs (8); B spectral branch — R_spec, +0.397%, MC p-values (5); C dust≡CDM code-consistency — residual 0, c_s^2=0.01 control 7.4% (3); D CLASS+HyRec — z_{rec}=1088.8, r_s=144.55, branch Δz=+0.40%, Δr_s=−0.26%, ΔH_0=+0.94, branch χ^2 3145→37→0.73 (6); E lab-vs-CMB — H1 vs lab −0.37%, H2 +0.02%, CMB 0.86σ subdominant, no-variation 1.8σ, AT-11 (5); F action→dust — conservation, linearity p=0, $T_{\mu\nu}$ form, C-dust-1a/1b/2/3/4 (5); G joint program + narrowed claim + outcomes A/B/C (4); H reproducibility — scripts released, CLASS settings stated (5); I (v1.4) executed joint test — ω_b/ω_c/n_s offsets, h fit to θ_s, $\Delta\chi^2$≈0 (noise) / ≈204 (CVL) (4); J (v1.4, revised v1.6 per ZS-M20) within-branch σ_{th} now NNLO-limited (σ-ratio term removed, 3477 EXACT); lab favors H2 (+0.020%) over H1 (−0.374%) by magnitude; 6.3σ/0.34σ retracted (4); K (v1.5) C1 Brown variational — $T^{\mu\nu}=\rho u^\mu u^\nu$ via held-fixed $\mathscr{J}^\mu$, p=0, conservation identity, FRW timelike $J^2=-n^2<0$, 32:6→16/3 under μ-univ. (4); L (v1.5) C2 ZHCS species in self-consistent Einstein-fluid solver — $\Delta\chi^2_{iso}$=0 vs CDM across k, imperfect/wrong-sign diverge (3); M (v1.5) C3 Planck-calibrated parameter-surrogate — Fisher ρ rescaled to Planck σ, profiled $\Delta\chi^2$=1.65 (3 dof, p=0.65), AT-6a PASS / AT-6b clik TARGET (3); N (v2.0) C1 continuum closure — parent-action Brown identities, $J^\mu=\partial_\nu \Sigma_Z^{\mu\nu}$ reduction to A19, single-trace TrP_c/TrP_b=16/3 (μ-universality eliminated), shared-clock S_{cb}=0 and reduced kinetic rank 1, C1-ID-global OPEN (17). All 76 PASS; COMPUTED/VERIFIED-IN-EINSTEIN-FLUID-SOLVER items reproducible from released scripts (compute_a20, class_a20_pathA, minicmb_a20, fisher_a20, continuum_a20); the custom-CLASS run and full clik MCMC are excluded as the two TARGETs.

References

[1] K. Kang, ZS-F2 / ZS-F5 (Z-Spin Collaboration, 2026): A = 35/437; (Z,X,Y)=(2,3,6), Q=11.
[2] K. Kang, ZS-M11 §9.5.7 (2026): Q-pair/X-pair decomposition; R_spec = (5−φ)/(4−φ).
[3] K. Kang, ZS-S8 §6.17; ZS-S9 Cor. I (2026): tau-mass H1/H2 branches; $m_e = m_\tau/3477$.
[4] K. Kang, ZS-A19 v3.1 (2026): boundary-tension cold dark matter; Ω_{cdm} = 32/121.
[5] S. Weinberg, Phys. Rev. D 67, 123504 (2003); arXiv:astro-ph/0302326.
[6] J. D. Brown, Class. Quantum Grav. 10, 1579 (1993); arXiv:gr-qc/9304026. (Action functionals for relativistic perfect fluids; densitized-current dust → $T^{\mu\nu} = \rho u^\mu u^\nu$, p = 0.)
[7] J. D. Brown and K. V. Kuchař, Phys. Rev. D 51, 5600 (1995); arXiv:gr-qc/9409001.
[8] C.-P. Ma and E. Bertschinger, Astrophys. J. 455, 7 (1995); arXiv:astro-ph/9506072.
[9] D. Blas, J. Lesgourgues, T. Tram, JCAP 07, 034 (2011); arXiv:1104.2933. (CLASS Einstein–Boltzmann code.)
[10] J. Torrado and A. Lewis, JCAP 05, 057 (2021); arXiv:2005.05290. (Cobaya sampler with clik likelihoods.)
[11] ACT Collaboration (T. Louis et al.), arXiv:2503.14452 (2025). (ACT DR6 CMB power spectra and likelihood.)
[12] DESI Collaboration, arXiv:2503.14738 (2025). (DESI DR2 BAO cosmology.)
[13] Y. Ali-Haïmoud and C. M. Hirata, Phys. Rev. D 83, 043513 (2011) (HyRec); N. Lee and Y. Ali-Haïmoud, Phys. Rev. D 102, 083517 (2020) (HYREC-2).
[14] D. Blas, J. Lesgourgues, T. Tram, JCAP 07, 034 (2011) (CLASS); arXiv:1104.2933.
[15] L. Hart and J. Chluba, Mon. Not. R. Astron. Soc. 493, 3255 (2020); arXiv:1912.03986.
[16] Planck Collaboration, Astron. Astrophys. 580, A22 (2015); arXiv:1406.7482 (varying constants).
[17] Y. Toda and O. Seto, arXiv:2508.09025 (2025): ACT DR6 + DESI DR2 varying-m_e, m_e/m_{e0} = 1.0081 ± 0.0046.
[18] ACT Collaboration, arXiv:2503.14454 (2025): P-ACT-LBS m_e/m_{e0} = 1.0063 ± 0.0056.
[19] N. Schöneberg and L. Vacher, JCAP 03, 004 (2025); arXiv:2407.16845 (the mass effect).

[20] P. J. Mohr, D. B. Newell, B. N. Taylor, Rev. Mod. Phys. 93, 025010 (2021) (CODATA 2018).
[21] Planck Collaboration, Astron. Astrophys. 641, A6 (2020); arXiv:1807.06209.

Version History

v2.0 (June 2026): Continuum-closure revision — implements the reviewer's parent-action program and specifies the two computational programs. (CONTINUUM, §9.1–§9.5) A first-order parent action $S = -\mu_Z\int\sqrt{(-g_{\mu\nu} J^\mu J^\nu)} + \int J^\mu\partial_\mu T$ is shown by variation to give Brown dust ($\partial_\mu J^\mu=0$; $u_\mu=\partial_\mu T/\mu_Z \Rightarrow$ irrotational; $T^{\mu\nu}=\rho u^\mu u^\nu$, $p=0$); writing $J^\mu=\partial_\nu\Sigma_Z^{\mu\nu}$ reduces it to the A19 antisymmetric-current form, establishing the bidirectional equivalence and DERIVING the Σ_Z→Brown-current identification that v1.6 only adopted (C1-ID-local). A single-trace matrix current with rank-32 and rank-6 orthogonal projectors makes $\rho_c/\rho_b = \mathrm{Tr}P_c/\mathrm{Tr}P_b = 16/3$ a structural identity, ELIMINATING μ-universality (C1-μ). A shared clock $J_i^\mu=N_i j^\mu$ gives $\delta_i=\delta n/n$, $\theta_i=\theta$, $S_{cb}=0$, and a reduced scalar kinetic matrix of rank one (single physical clock, ghost-free, $c_s^2=0$), DERIVING single-clock adiabaticity (C1-clock, C1-entropy). Net: the A19 boundary action → linear-cosmological dust is promoted from DERIVED-CONDITIONAL on three conditions (v1.6) to DERIVED, conditional only on the global charge quantization C1-ID-global (N_i=32,6 as integer $H^3(M_Z,\mathbb{Z})$ fluxes), the sole residual; new continuum gate table (Table 3c) and verification category N (continuum_a20.py, 17 checks). (BOLTZMANN PROGRAM, §8.5) Gives the exact custom-CLASS species patch (input/background/perturbation modules, Newtonian and synchronous gauges, adiabatic ICs, outputs) and the equivalence gates B1–B6 (Table 2b); the compiled in-CLASS run that would raise VERIFIED-IN-EINSTEIN-FLUID-SOLVER to VERIFIED-IN-FULL-BOLTZMANN remains TARGET. (LIKELIHOOD PROGRAM, §11.2) Pre-registers model ZS-C (geometry-restricted, zero new shape parameters; sample h,τ,A_s+nuisance) vs the aspirational strict ZS-0, the run plan P0–P2/A1–A3/D1, the data-level gates L0–L10 (Table 5b), a mock-calibrated empirical-p / ΔAIC decision rule for AT-6b, and a data-provenance discipline against double-counting; AT-6a still PASSES. Adds CLASS, Cobaya, ACT DR6, DESI DR2 references. All v1.6/v1.5/v1.4 content retained; abstract updated; ZS-M20 σ_1/σ_3=3477 EXACT and the magnitude-based H2 branch carried unchanged. Verification 76/76 (92 sub-checks in released script). The two TARGET-COMPUTATIONs are now the custom-CLASS run and the data-level likelihood; the continuum's only open lemma is C1-ID-global.
v1.6 (June 2026): Tightening / corpus-consistency revision — responds to the v1.5 review on three scope phrasings and one corpus conflict. (CRITICAL, §10.1) ADOPTS ZS-M20: $\sigma_1/\sigma_3 = (Q+Y)^2\cdot G + (Q-Z) = 3477$ EXACT (G=12; The Book §M17.4 LOCKED) supersedes the older ZS-M11 §5.2 value 3475, so the formerly dominant 0.0575% σ-ratio term is REMOVED from the within-branch m_e error budget; σ_{th} becomes NNLO-limited (OPEN), and the v1.4/v1.5 significances 6.3σ (H1) / 0.34σ (H2) are RETRACTED. The branch is now selected by magnitude (lab m_e matches H2/R_spec to +0.020% and misses H1/√2 by −0.374%), with the sharp prediction that the NNLO lepton-Yukawa Schur correction must be ≲ 0.02%; Figure 2b and Table 4b are revised accordingly. (§9, title) The A19-action→dust link is restated as DERIVED-CONDITIONAL on THREE named conditions — adding the Σ_Z→Brown-densitized-current identification (adopted, not proven) to μ-universality and single-clock — and the title is softened from "Closing" to "Tightening" the continuum chain. (§8.4) The solver is relabeled VERIFIED-IN-EINSTEIN-FLUID-SOLVER (radiation+dust+one potential, not a full Boltzmann hierarchy), with the honest notes that identical equations+ICs make ZHCS=CDM expected by construction and that a genuine C2 closure needs a custom in-CLASS zhcs species. (§11.1) The likelihood is relabeled a Planck-calibrated Gaussian parameter-surrogate and AT-6 is split into AT-6a (surrogate, PASS) and AT-6b (actual clik, TARGET); the real covariance is noted as extractable from public Planck chains. "Zero free parameters" → "zero new fitted Z-Spin shape parameters" (h profiled to θ_s; τ, A_s marginalized). Editorial: abstract trimmed to ~300 words and version tense fixed; verification count unified (59 itemized; 75 sub-checks in zs_a20_verify_v1_6.py); Code Availability and Appendix A.2/A.3 now document minicmb_a20.py, fisher_a20.py and class_a20_pathA.py (equations, ICs, k-grid, tolerances, Fisher steps, Planck σ source, covariance, $\Delta\chi^2$=1.65 reproduction); Table 3/Table 5 rows set not to split across pages. All v1.5/v1.4 content retained. Verification 59/59. The full clik MCMC and a custom in-CLASS zhcs species remain the two TARGET-COMPUTATIONs.
v1.5 (June 2026): Gate-closure revision — closes the three items the prior review left open. (C1, §9) CONTINUUM CLOSURE: replaces the assumed stress tensor with the Brown (1993, gr-qc/9304026) densitized-current variational derivation — holding $\mathscr{J}^\mu = \sqrt{-g} J^\mu$ fixed under δg gives $T^{\mu\nu} = \rho u^\mu u^\nu$, p = 0 exactly (variational ambiguity closed). C-dust-1a (conservation) and C-dust-2 (timelike on FRW, $J^2=-n^2<0$) are upgraded to DERIVED; C-dust-4 to DERIVED (scalar sector; vector modes decay); stability/ghost-freedom PROVEN (imported). C-dust-1b (32:6→ρ_c/ρ_b=16/3) and adiabatic occupation become DERIVED-CONDITIONAL on two named assumptions (μ-universality; single Z-sector clock). Net: action→dust upgraded from DERIVED-CONDITIONAL to DERIVED-CONDITIONAL on three named conditions. (C2, §8.4) BOLTZMANN SPECIES: evolves the ZHCS dust as its own species in a self-consistent Newtonian-gauge Einstein-fluid integrator (Φ sourced by all

species, genuine back-reaction — not a fixed-Φ toy); it reproduces standard CDM to machine precision ($\Delta\chi^2$_iso = 0, k = 10^{-3}–0.3 Mpc^{-1}) while imperfect (c_s^2=0.01) and wrong-sign forms diverge, upgrading the Boltzmann Isomorphism to PROVEN-analytic + VERIFIED-IN-EINSTEIN-FLUID-SOLVER (a fluid-level, not full-Boltzmann, check). (C3, §11.1) LIKELIHOOD: replaces the spectrum-level CVL proxy with a Planck-calibrated parameter likelihood — a CLASS-derivative Fisher correlation matrix rescaled to the published Planck 2018 1σ errors; the zero-parameter point gives $\Delta\chi^2$ = 1.65 (3 dof, p = 0.65), so AT-6a PASSES (parameter-surrogate; 1D pulls ω_b +1.2σ, ω_c +0.21σ, n_s +0.60σ). Adds Figure 3 and the Brown 1993 reference. All v1.4/v1.3 content retained. Verification 59/59. Zero free parameters; the full clik/Cobaya MCMC over raw data with foreground nuisance remains the one TARGET-COMPUTATION.

v1.4 (June 2026): Protocol-to-result revision. (1) EXECUTES the zero-parameter joint test (§8.3): the Z-Spin spectrum (absolute lock Ω_c = 32/121, Ω_b = 6/121, ω_i = Ω_i h^2, n_s = 0.9674, time-invariant m_e, h fixed by θ_s) vs a Planck-2018 proxy gives $\Delta\chi^2 \approx 0$ at Planck noise (outcome A, PASS) and ≈ 204 cosmic-variance-limited (future discriminator), with Figure 2 — the first Z-Spin spectrum confronted with data with no fitted shape parameter. (2) DERIVES the within-branch electron-mass theory error σ_th = 0.060% (dominated by the DERIVED σ-ratio chain 3475-vs-3477, §10.1), under which H1 is EXCLUDED at 6.3σ and H2 CONFIRMED at 0.34σ, resolving the prior review's inconsistency (0.4% is the branch separation, not the within-branch error) and yielding a sharp falsifiable prediction (a future NNLO Schur computation must reproduce m_e to ≤ 0.06%). (3) RECLASSIFIES the v1.3 H1/H2 varying_me spectra as COUNTERFACTUAL-SENSITIVITY (they used a z=50 transition, a time-varying model Z-Spin does not predict); the genuine Z-Spin input is varying_me = 1, used in §8.3. Corrects the Ma–Bertschinger sign (δ' = −θ + 3Φ'); splits C-dust-1 into 1a (conservation, DERIVED) / 1b (32:6 norm, OPEN) and adds C-dust-4 (vorticity, OPEN). Enforces ω_i = Ω_i h^2 consistency in §8.3 (absolute Ω lock). All v1.3 content retained. Verification 44/44. Zero free parameters; full clik MCMC remains TARGET-COMPUTATION.

v1.3 (June 2026): Executed-computation revision. Runs CLASS+HyRec (real recombination z_rec = 1088.8, replacing the spurious Saha 1378) for fiducial/H1/H2 and confirms the branch differentials Δz_rec = +0.40%, Δr_s = −0.26%, ΔH_0 = +0.94 km/s/Mpc; produces real lensed TT/TE/EE spectra (Figure 1). Derives the perfect-dust stress tensor from the A19 BTD-3 boundary action (§9), upgrading the action→dust link to DERIVED-CONDITIONAL with three residual conditions (C-dust-1/2/3). CORRECTS the central v1.2 claim: the laboratory electron mass (ppb) — not the CMB (0.86σ) — selects the branch (H2; H1 excluded); v1.2 "cosmology selects the branch" RETRACTED (gate AT-11). Relabels the v1.2 Fisher $\Delta\chi^2$ = 6.95 (σ = 0.15%) as ILLUSTRATIVE-FORECAST; updates current sensitivity to ACT DR6+DESI DR2 (0.46%, 0.86σ). Narrows "no analogue" to the specific joint spectral-m_e × geometry-fixed-abundance test. Narrows dust≡CDM to COMPUTED code-consistency under perfect-dust antecedents (full ODE units/ICs/tolerances stated). Releases companion scripts (§6). Fixes the v1.2 Acknowledgements copy-paste; removes the empty auto-TOC and trailing blank page; stabilizes sub/superscript typography. Verification 36/36. Zero new fitted parameters; clik MCMC remains TARGET-COMPUTATION.

v1.2 (June 2026): Added Saha/Friedmann differentials, dust≡CDM ODE, illustrative Fisher, and the joint {branch, abundance} program. v1.1 (June 2026): spectral origin of H2, status corrections (Weinberg existence/occupation+C3, Ω/ω, AT-3), Word styles. v1.0 (June 2026): reframed the recombination floor; partially RETRACTED the near-NO-GO; imported Weinberg 2003 and Brown–Kuchař 1995.